Instruction cycle (IC) → Time a µp req. to complete FDE cycle of a single instruction

IC = Fetch cycle (FC) + Exe. cycle (EC)

↓ c/o one or more machine cycles/bus cycles

↓ time req. to carry out a simple read/write operation of a byte from mem ↔ I/O

↓ c/o 4/more clock periods (T1, T2, T3, T4)

1 IC = 1/more machine or bus cycle

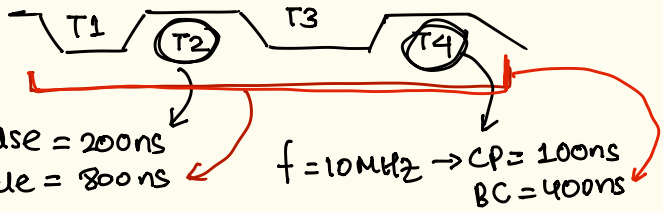
= 4 clock periods (T1–T4)

8086 µp → req. ext. clock cyc. generator (8284A)

↳ connected to pin 19 (CLK) of 8086

5–10 MHz freq. → higher the freq., lower the clock pulse, faster exe. rate

$T = \frac{1}{f}$



f = 5 MHz

↳ Clock pulse = 200 ns

bus cycle = 800 ns

f = 10 MHz → CP = 100 ns

BC = 400 ns

Duty cycle → % of time a signal is ON (active) during one complete cycle.

$$\text{Duty cycle} = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100\%$$

↓

For 8086 µp, usually duty cycle = 33%

TIMING IN 8086

Each bus cycle ≡ 4 T-state

T1:
- addr. sent out via addr. bus
  - mem loc.
  - I/O
- ALE, DT/$\bar{R}$, M/$\bar{IO}$ are also output
  - ALE → addr. bits in AD0-15
  - DT/$\bar{R}$ → DT=1 (write), $\bar{R}$=0 (read)
  - M/$\bar{IO}$ → M=1 (interact^n w mem), $\bar{IO}$=0 (" " IO)

T2:
- 8086 issues RD or $\bar{WR}$, $\bar{DEN}$ signal
  - $\bar{WR}$=0 (data to be written appear on data bus)
- $\bar{DEN}$ signal → turns on data bus buffers (if they are present in the system)
  - so that mem/IO can receive data to be written (WRITE)
  - OR
  - up can accept data to be read from mem/IO (READ)
- If write bus cycle → data sent out to mem/IO through data bus
- READY → sampled at the end of T2.
  - If READY=0 → T3 becomes (T_w) → clocking period allowing mem. time to access data.

NOTE: ① wait state (T_w) → extra clocking period inserted bet^n T2 and T3.

② sampling → Say " sampling the data bus " → up reads & latches (captures) the binary data present on the data bus at a specific instant in time when the data is guaranteed to be valid.

T3:
- If bus cycle = READ bus cycle → data bus sampled at the end of T3

T4: system
- RESET → all bus signals deactivated in prep for the next bus cycle.

Summary

T1: addr. (addr. set/) generated by up
T2: up generates signal to indicate what type of operation it will perform (M/IO, RD/WR) [Bus cycle]
T3: Data transfer/receive start
T4: Ctrl signal removal/ reset system

Maths

① Duty cycle 60% & for each clock pulse assume $T_{OFF} = 40\,ns$. Frequency = ?

$T_{ON} = 0.6T$, $T_{OFF} = 0.4T$

Given, $T_{OFF} = 0.4T = 40\,ns$

$\Rightarrow T = 100\,ns$.

$\therefore$ Frequency, $f = \frac{1}{T} = 10MHz$

② 8086 clock $f = 8MHz$

READ operation : MOV AX, [1234h]



Req. 1 BC (from memory bank concept)

$\therefore$ IC time = 1 BC = 4 or more clock cycle (अगर no other info given, so 4 cc)

$\because$ 1 IC req 1/more BC

$1CC = T = \frac{1}{F} = \frac{1}{8 \times 10^6} = 125\,ns.$

ⓐ Total time req. = ?

ⓐ Tot time req = 1 IC = $4 \times 1cc = 4 \times 125ns = 500ns$

ⓑ Think: T_{ON} = ?

T_{OFF} = ?

③ 8086 → f = 5MHz

3 instructions being executed:

i) MOV AL, [1000h] ⟶ 1 bus cycle ⟶ no waiting time

ii) MOV BL, [2000h] ⟶ 1 " " ⟶ 2 waiting time

iii) MOV CL, [3000h] ⟶ 1 " " ⟶ 4 waiting time

Total time req = ?

For (i) $1\,BC = 4CC = 4 \times \frac{1}{5\times10^6} = 4 \times 200\,ns = 800\,ns$

(ii) $1BC = 4CC + 2WC = 6CC = 6 \times 200\,ns = 1200\,ns$

(iii) $1BC = 4CC + 4WC = 8CC = 8 \times 200\,ns = 1600\,ns.$

Total IC = 800 + 1200 + 1600 = 3600 ns

④ Total BC = 500ns, no. of T-state = 4 with no wait cycle

ⓐ f = ? ⓑ clock period time = ?

ⓒ if 2 waiting state introduced & duty cycle = 50%, the 1 bus cycle remains ON for? → T_{ON} = ?

ⓐ $1\,BC = 4T \Rightarrow T = \frac{500\,ns}{4} = 125\,ns.$

$f = \frac{1}{T} = \frac{1}{125} = 8MHz$

ⓑ Clock period time = T = 125ns

ⓒ Total cycle = 4 T-state + 2 wait state
= 6 clock cycle

∴ 1 BC req. = 6 × 125 ns = 750 ns

$T_{ON} = \frac{50}{100} \times 750$